

Clinical Pharmacokinetics: Rational Dosing and the Time Course of Drug Action

Practice Quiz Questions, Answers, and Rationales (Explanations)

Pharmacology bridges physiology, pathophysiology, and clinical practice.

Some items may have more than one answer.

CHALLENGE: Cover the options and try to answer the questions without looking at the choices. Then, explain why the other options are incorrect.

Goals of Practice Quizzes:

1. To help students identify what they know and understand by applying knowledge to clinical case vignettes.
2. To help students get used to thinking through the case vignettes that are used “Board-style” questions. NYITYCOM, COMLEX and USMLE examine students’ medical knowledge using these types of questions.

My recommendation is to highlight in the vignette the relevant information (“clues”) you need to answer the question. *Don’t let yourself get distracted by the background information unless it is relevant for answering the question.*

Where can I find the answers to the questions?

If you wish to use them, all answers can be found by **doing a “search” of the Notes Handout** or the Examination and Board Review resources designated on the CPG (links are provided). The handout and textbook readings are not a required, but you may find one or the other to be VERY helpful and relevant.

THE ANSWER KEY WILL BE POSTED Nov. 21, 2025.

For multiple-choice questions, explain why the incorrect options are not correct. In all cases, explaining the answers (out loud) will help solidify the learning.

First, the basics

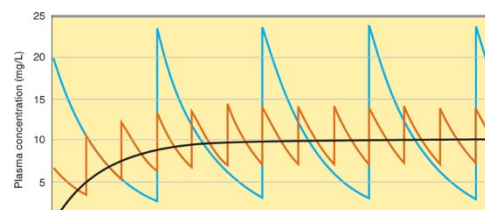
1. Rational dosing applies the dose-effect relationship. What are two goals of this fundamental approach to safe drug therapy for the individual patient?

Hint: Perhaps the figure will give you a clue.

Short answer.

Rationale: The therapeutic goals are to maintain steady state within the therapeutic window. That is:

- 1) To maintain the desired therapeutic effect



Basic and Clinical Pharmacology Figure 3-6

Source: Bertram G. Katzung:
Basic & Clinical Pharmacology, Fourteenth Edition
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2) with minimal adverse effects.

Dose-effect relationship: Knowing the relationship between dose, drug concentration, and effects allows the clinician to take into account the various pathologic and physiologic features of a particular patient that make that patient different from the average individual in responding to a drug.

The importance of pharmacokinetics and pharmacodynamics in patient care thus rests upon the improvement in therapeutic benefit and reduction in toxicity that can be achieved ***by application of these principles***. (Katzung, 2023; my bold and italics)

Legend on Katzung Figure 3-6: Relationship between frequency of dosing and maximum and minimum plasma concentrations when a steady-state theophylline plasma level of 10 mg/L is desired. The smoothly rising black line shows the plasma concentration achieved with an intravenous infusion of 28 mg/h. The doses for 8-hour administration (orange line) are 224 mg; for 24-hour administration (blue line), 672 mg. In each of the three cases, the mean steady-state plasma concentration is 10 mg/L.

2. What are the 4 most important parameters governing the concentration, movement, and time course of drug effect in the body, collectively known as drug disposition?

Hint: Perhaps the simple model at the right will help you. (Notes p1)

Short answer

Rationale:

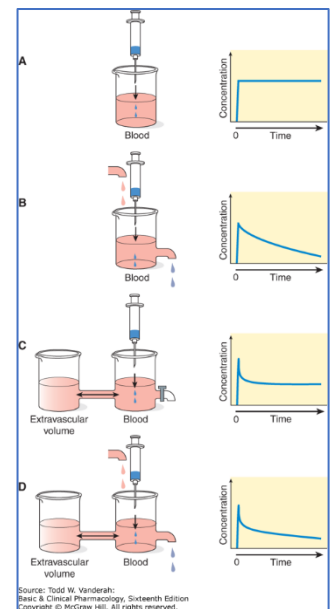
Clearance, bioavailability, volume of distribution, and elimination half-life

- **Clearance** is an important parameter in determining a rational maintenance therapeutic regimen.

The goal of therapy is to maintain the ***steady state concentration*** within the therapeutic window, which is achieved when ***the rate of drug elimination (clearance) equals the rate of drug administration***.

- **Bioavailability** describes the fraction of the drug dose that reaches the systemic circulation, which determines the concentration of drug at the effector site.
- **Volume of distribution** (apparent) is the ratio of the amount of drug in the body (the dose) to the concentration of drug measured plasma (or blood). V_d represents the degree to which a drug is distributed in body tissue – it is directly proportional to the amount of drug distributed to tissue.

The higher the V_d , the greater the amount of tissue distribution.



Basic & Clinical Pharmacology
Figure 3-2

Recall the principle: A relationship exist between a drug's effect (beneficial or toxic) and the concentration of drug that reaches the effector site.

- **Half-life** is the time required to change the amount of drug in the body by one-half.

It is a *proportional* decrease in drug concentration.

As drug concentration decreases, the rate of elimination slows down proportionately.

Half-life is useful because it shows the time course of drug accumulation during a constant rate of drug administration and the time course of drug elimination after stopping drug administration.

3. A patient is given an intravenous bolus of a drug that achieves a peak serum concentration of 1.75 mcg/mL. The plasma concentration vs. time curve shows a steep decline initially, then a more gradual decline.

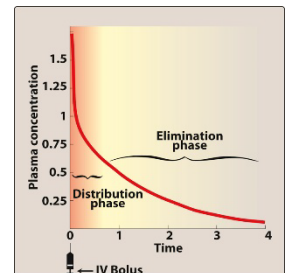
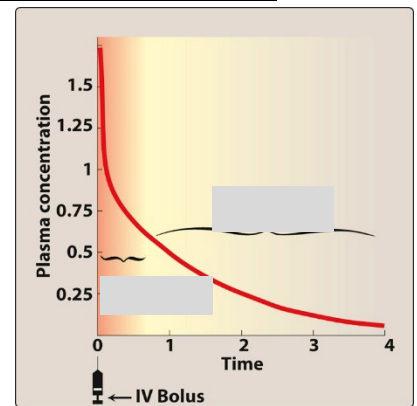
- i) What is the explanation for the biphasic concentration-time curve?
- ii) What are the 2 phases of the curve called?

Rationale:

- **Early distribution phase** shows a rapid decline (steep negative slope) as the drug leaves the plasma and distributes to the interstitial fluid and tissues.
- **Slower elimination phase** shows elimination (more gradual negative slope) of the drug from the body corresponding to the drug's elimination kinetics.

❖ **Pearls:**

- The elimination phase occurs after distribution once the drug has reached a relatively stable concentration between plasma and tissues.
- Elimination is a continuous process that may overlap with distribution.
- The half-life can be estimated from the terminal slope of the plasma log concentration-time curve.



4. A 37-year-old woman with major depressive disorder is started on a standard low dose of an antidepressant. The doctor explains that the antidepressant effect usually take several weeks. Therapy will be started with a low dose that will be gradually increased to optimal therapeutic effect based on response and on tolerability. The drug has a $t_{1/2}$ of 7 days.

- i) What is the approximate time to steady state concentration?
 - A. 7 days
 - B. 14 days
 - C. 28 days*
 - D. 70 days
- ii) How often should a dosage increase be considered, if needed?
 - E. 7 days

- F. 14 days
- G. 28 days*
- H. 70 days

Please explain your reasoning.

Rationale: C and G: 28 days

i) Steady state concentration is approached in 4 half-lives ~94% and 5 half-lives ~97%.

For a drug with a 7-day half-life, steady state is achieved in 28 to 35 days.

The 4 half-lives time frame is *typically* associated with a full therapeutic response

— assuming the dose is correct and the patient is responsive —

ii) *As a general statement*, the dose would be increased in about 4 half-lives to allow time to reach the new steady state between dose increases.

❖ **Pearls:**

In the example of the antidepressant in the example, dose adjustment may occur between 3 and 4 weeks depending on urgency and patient response.

A low initial dose may not attain the optimal therapeutic effect or even the minimal effective concentration at steady state – depending on patient variables.

- Patient-specific factors affect response, such as age, weight, metabolism, comorbidities, renal and hepatic function
- Narrow therapeutic window: Cautious increases closer to steady state, 4 half-lives, to reduce the risk of toxicity
- Monitor! during dose adjustments according to recommendations listed in the monograph

➤ **Practice point:** Dosage adjustments should be done using *rules of thumb*:

- Change the dosage by no more than 50% and no more often than every three to four half-lives to assure that near steady state concentration has been achieved.
- Adjusting the dose more often could lead to an overshoot of the desired target dose based on the principle of the time course of drug accumulation and drug elimination during constant-rate drug administration.
- For drugs with defined therapeutic range, dose and frequency can be adjusted to obtain desired plasma levels.

5. A 76-year-old man is hospitalized for the management of acute ischemic stroke and blood pressure of 195/112 mmHg after reperfusion therapy. The patient is given an IV loading dose of labetalol (a beta blocker) followed by continuous intravenous infusion of the maintenance dose of the drug to reduce his blood pressure to a target of <180/105 mmHg in the first 24 hours. Labetalol has a half-life of about 5.5 hours, according to information in the drug monograph. What is the purpose of this dosing strategy?

Rationale:

Dosing strategy: a loading dose followed by maintenance dose

Loading dose is administered to rapidly bring drug concentration up to the therapeutic level.

A loading dose is administered as a single large dose, or smaller doses over a relatively short time period.

When is a loading dose administered?

- ... When a patient requires a therapeutic concentration immediately – even drugs with relatively short half-lives: antibiotics, heparin, antiseizure (status epilepticus)
- ... Drugs that have long half-lives and the therapeutic effect should not be delayed

When is a loading dose not administered?

- ... When a patient's condition allows for a gradual onset – many stable, chronic conditions
- ... When a drug is associated with significant toxicities – start low, go slow is a safer strategy – tolerance to common side effects can develop with gradual increase in blood levels
- ... When the loading dose would cause dangerously high drug levels

Maintenance dose maintains the levels in the therapeutic window over time as drug is lost through elimination (metabolism and excretion).

A maintenance dose is a smaller dose administered in regular intervals or by continuous infusion.

When is the maintenance dose administered:

It may be started immediately after the loading dose or delayed depending on:

- ... Urgency of the clinical condition (cardiac arrest, status epilepticus),
- ... Patient factors that influence elimination (renal / liver function, age, weight), and
- ... A delay may be necessary for the drug to distribute to minimize toxicity.

MONITOR

Therapeutic drug levels, when appropriate
Signs and symptoms of response
Signs and symptoms of toxicity

Calculations:

$$LD = \frac{V_d \times C_p \text{ desired}}{F}$$

$$MD = \frac{CL \times C_p \text{ desired}}{F}$$

The drug in the vignette is administered IV so bioavailability $F = 1$.

When calculating LD and MD for IV drugs, $F=1$ is frequently omitted from the equation.

Why are V_d and C_p the loading dose parameters?

Volume of distribution and desired plasma concentration take into account the initial distribution of drug into the tissues.

- The loading dose rapidly brings the plasma concentration to a desired target, $C_{p \text{ desired}}$ – which is the steady state concentration, C_{ss} .

$V_d = V_{ss}$, the volume of distribution at steady state, which is clinically relevant. As the drug enters the systemic circulation, it distributes from the bloodstream to the tissues and compartments of the body. Therefore, the loading dose must account for the amount of drug

- “lost” to the tissues.

Why are CL and C_p the maintenance dose parameters?

Clearance and desired plasma concentration take into account that the drug has distributed to the tissues and a stable equilibrium between blood tissues has been established.

Steady state: The rate of drug going out (CL) = the rate of drug going in (C_p)

The point at which the amount of drug being eliminated equals the amount of drug being administered = equilibrium

❖ Remember the fundamental principle of steady state:

The loading dose delivers a therapeutic concentration but it still takes 4 half-lives to approach steady state concentrations. Starting the maintenance dose shortly after the loading dose maintains the drug concentration in the therapeutic range.

Monitoring

The dose would be adjusted according to blood pressure response.

6. The patient in the above vignette is given labetalol continuous infusion at 5 mg/minute. The dosing rate calculation included the value 105 L/h for a 70 kg adult obtained from a table. This parameter is defined as the volume of blood plasma (105 liters) removed of drug by elimination over a specific time period (1 hour) in a 70 kg adult.

Which of the following is defined by this parameter?

- A. Area under the concentration-time curve
- B. Clearance*
- C. Steady state
- D. Volume of distribution

Define each term.

Rationale: A. Clearance

Please see the table on the next page. Be sure you can define each term, explain the concepts and the reasons why they are crucial to patient care.

Clearance (CL)

Clearance determines how efficiently the body eliminates a particular drug

The **proportionality factor** that relates the **rate** of drug elimination to the **concentration** of drug in plasma

- The rate of elimination is the fraction of drug eliminated over time – the rate at which plasma concentration will decline during the elimination phase.

Thus, clearance is the volume of blood cleared of drug per unit of time – 105 L/h/70 kg in the case of labetalol (value from Katzung Table 3-1)

Bioavailability (F)

Drug levels in the systemic circulation are influenced by absorption and first-pass effect

The fraction (*F*) of the administered dose that reaches the systemic circulation

Volume of distribution (Vd)

Reflects the extent to which the drug stays in plasma vs distribution to tissues

The *apparent* (theoretical) space in the body available to contain the drug. It does not necessarily refer to an identifiable physiologic volume.

- V_d relates the amount of drug in the body (dose) to the concentration of drug measured in plasma or blood → $V_d = \text{Dose in mg} / C_p \text{ in mg/L}$

Half-life ($t_{1/2}$)

Determines how long a drug stays in the body, time to steady state, and guides dosing intervals

The time it takes for drug levels to reduce by one half by elimination

Area under the curve (AUC)

Represents the total exposure of the body to the drug over time

The measured concentration of drug over time (from zero to infinity). AUC: mg*h/L

Steady state

A dynamic equilibrium where the drug concentration remains relatively constant over time

The rate at which the drug administration equals the drug elimination

Drug concentration fluctuates within a predictable range

Achieving steady state is crucial to ensure the drug's effectiveness while minimizing the risk of adverse effects / toxicity.

7. A critically ill patient is administered the antibiotic gentamicin for a severe gram-negative infection. Dosing calculation included the V_d value 0.25 L/kg.

What does this value indicate about the drug's distribution?

Short answer

Hint: Notes p10

Rationale: The drug stays mainly in body water

0.25 L/kg is a very low apparent volume of distribution.

Drugs with very low V_d tend to remain in the extracellular compartment.

Thinking it through:

- An average 70 kg male has ~42 L of body water – ~60% of his weight.
- About 2/3 (28 L) is within the intracellular space and 1/3 (14 L) is extracellular fluid volume.
- The extracellular compartment consists of body water – plasma and interstitial fluid.
- Saying, “Drugs that remain in the extracellular compartment” infers that they are hydrophilic and, therefore, do not cross physiologic barriers to the intracellular compartments of tissues.
- Thus, a low Vd indicates low distribution to body compartments – The drug stays in the extracellular fluid volume.

Conversely, lipophilic drugs that distribute throughout cells, muscle, and adipose tissue have very large volumes of distribution.

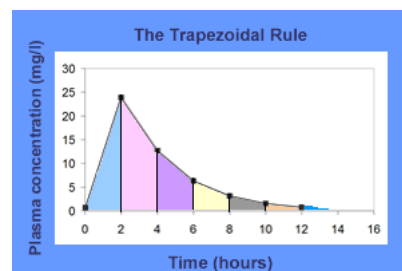
8. A normally healthy 56-year-old man is hospitalized for treatment of a resistant staphylococcal (MRSA) infection with I.V. vancomycin (antibiotic). Plasma sampling is drawn for AUC-based dosing.

Which of the following relationships BEST reflects this parameter selected for optimizing vancomycin dosing in this patient?

- The dose in relation to the plasma concentration
- The plasma concentration in relation to the blood flow
- The rate of drug elimination in relation to plasma concentration
- The time required to reduce the plasma concentration by one-half
- The variation in plasma concentration over time*

Rationale: AUC assesses the extent of total body exposure to the drug by integrating variation in plasma drug concentration over time after a single dose or during a single dosing interval, irrespective of the rate of absorption.

For vancomycin, the AUC-based protocol guides dosing, reduces risk of treatment failure, and reduces the risk of nephrotoxicity.



Incorrect options:

- A. The dose in relation to the plasma concentration models volume of distribution, $V_d = \text{Dose}/C_p$.
- B. The plasma concentration in relation to the blood flow is considered in relation to extraction ratio, E. Blood flow (Q) determines the rate of presentation to the organ (liver or kidney) and the rate of exit from the organ.
- C. The rate of drug elimination in relation to plasma concentration is the definition of clearance.
- D. The time required to reduce the plasma concentration by one-half defines half-life.

9. A 65-year-old woman is brought to the emergency department because she experienced a seizure at home. History of seizure disorder is negative. Medical history is significant for type 2 diabetes mellitus and chronic kidney disease. She receives an intravenous infusion of the antiseizure drug levetiracetam.

Levetiracetam properties:

- V_d : ~0.5 L/kg
- plasma protein binding <10% (minimal)
- Not appreciably metabolized
- Excreted in urine as unchanged drug

Which of the following options should be anticipated with the use of this drug in this patient?

	Half-life	Dosage
A.	increased	increased
B.	decreased	decreased
C.	increased	no change
*D.	increased	decreased
E.	no change	no change

Definition – chronic kidney disease: Reduced ability of the kidney to filter the blood efficiently
glomerular filtration rate → reduced glomerular filtration rate

Rationale: D. Increased half-life requiring a lower dosage

$$t_{1/2} = \frac{0.693 \times V_d}{CL}$$

- Half-life is inversely proportional to clearance. As clearance decreases, the half-life increases.
- Clearance is defined as the rate of drug elimination from a volume of plasma per unit of time.
- As the rate of drug elimination increases (it takes longer for the body to remove the drug from plasma → $t_{1/2}$), the clearance decreases.
- 0.693 is a fundamental constant in the exponential decay process. It is derived from the natural logarithm of 2.

Thinking through the vignette:

- Levetiracetam has first-order kinetics.
- Levetiracetam is excreted in the urine as unchanged drug – meaning, it is active drug.
- The patient has reduced renal function. Thus, clearance of levetiracetam in urine will be reduced and the half-life will be increased.
- Reduced rate of excretion would lead to accumulation of the drug in plasma. Thus, plasma concentrations would be elevated.
- Thus, the dosage would need to be reduced because of the reduced clearance.

Renal dosing

10. A 64-year-old female patient (60 kg) is treated with experimental Drug A for type 2 diabetes. Drug A is available as tablets with an oral bioavailability of 90%. If the V_d at steady state is 2 L/kg and the desired steady-state plasma concentration is 3.0 mg/L, which of the following is the most appropriate oral loading dose of Drug A?

- A. 6 mg
- B. 6.66 mg
- C. 108 mg
- D. 360 mg
- E. 400 mg*

Rationale: Loading dose = $\frac{V_d \times C_{p \text{ desired}}}{F}$

$$LD = \frac{(2 \text{ L/kg} \times 60 \text{ kg}) \times 3.0 \text{ mg/L}}{0.9} = 400 \text{ mg}$$

11. A 45-year-old patient with small cell lung cancer has elected to participate in the trial of a new chemotherapy drug. It is given by continuous IV infusion at a rate of 10 mg/h. Plasma concentrations (C_p) are measured. Results are on the table at the right.

Time after start of the infusion	C_p in mg/L
1	0.7
4	3.0
8	3.6
16	3.84
24	4.0
30	4.0

What conclusion can be drawn from these findings?

- A. Volume of distribution is 30 L.
- B. Clearance is 2.0 L/h.
- C. Elimination follows zero-order kinetics.
- D. Half-life is 2 hours.*
- E. Doubling the rate of infusion would result in plasma concentration of 16 mg/L at 40 hours.

Rationale: half-life = 2 h

Thinking it through:

Starting with half-life:

- The C_p remains at 4.0 mg/L at 24 hours and at 30.
- Therefore, the steady state concentration is 4.0 mg/L.
- Therefore, the concentration would be 2.0 mg/L at one half-life. The time of this measurement is not listed on chart.
- Two half-lives, 75% of the final steady-state concentration, is 3.0 mg/L, which was reached at 4 hours.
- 4 hours = 2 half-lives, so 1 half-life = 2 hours.

Incorrect options:

Volume of distribution:

- Calculating from the half-life equation: $t_{1/2} = (0.693 \times V_d) / CL$,
- $V_d = (t_{1/2} \times CL) / 0.693 = (2 \text{ h} \times 2.5 \text{ L/h}) / 0.693 = 7.2 \text{ L}$
- Therefore, Option A is incorrect.

Clearance:

- Rearranging the maintenance dose equation, $MD = CL \times C_{ss}$,
 $CL = \text{Dosing rate} / C_p = 10 \text{ mg/h} / 4.0 \text{ mg/L} = 2.5 \text{ L/h}$
- Therefore, Option B is incorrect.

Elimination

- The exponential elimination indicates that the drug is eliminated by first-order kinetics.
Exponential: The rate of decrease is proportional to the remaining quantity

Doubling the infusion rate

- C_{ss} is 4.0 mg/L as seen above. With linear elimination kinetics, doubling the infusion rate (doubling the dose) to 20 mg/h would double the steady state plasma concentration to 8 mg/L, not 16 mg/L.

Source: Access Medicine Katzung & Trevor's Pharmacology Examination & Board Review, 14e, 2024; Chapter 3: Pharmacokinetics, Study Question #4.

12. Recall questions:

i) Which one of the following equations may be useful for modeling the effect of a diseased metabolizing organ on a drug's clearance?

- A. $E = CL / Q^*$
- B. $CL = (0.693 \times V_d) / t_{1/2}$
- C. $F = AUC (\text{oral}) / AUC (\text{I.V.})$
- D. $K_a = 1 / K_d$

ii) What do the other options represent.

Rationale: $E = CL/Q$

Extraction ratio is a clearance model of an organ's relative *intrinsic capacity* for eliminating a given drug from the systemic circulation over a single pass through the organ at steady state.

Extraction ratio refers to any drug administered by any route.

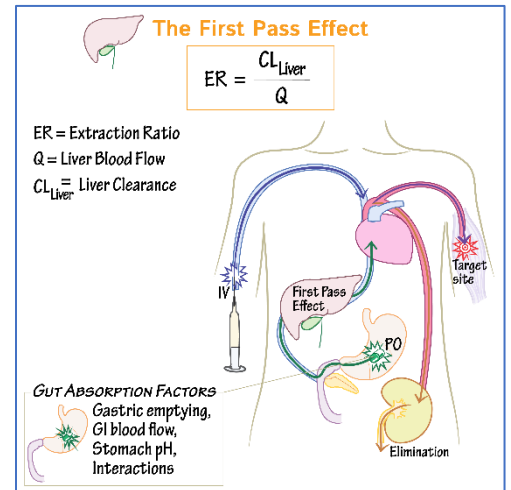
Calculations of extraction ratio, E , are useful for *modeling the effects of disease on clearance* of a drug by the clearance organs, especially hepatic and renal elimination.

The question indicated "a diseased metabolizing organ".

A critical function of the liver is metabolism and detoxification of xenobiotics – drugs, in pharmacology.

If the activities of the drug metabolizing enzymes because the function of a diseased liver is reduced, the plasma concentrations of drugs that are extracted by the liver can be significantly elevated.

- The figure on the right illustrates that an intravenous drug passes into the venous circulation entering the heart, followed by arterial transport to the target. It will eventually pass through the liver.
- Clearance of the drug is related to the liver's intrinsic capacity to eliminate it over a single pass through the liver at steady state.
- The figure also illustrates hepatic extraction of a drug administered orally. Since the drug passes through the enterocytes and liver before reaching the systemic circulation, it is called first-pass effect.



<https://www.drawittoknowit.com/course/pharmacology/glossary/pharmacology/extraction-ratio-the-first-pass-effect>

Explanation of the extraction ratio model:

The intrinsic capacity of the liver to metabolize a given drug to inactive metabolites can be determined by measuring the concentration of free (unbound) drug in the portal circulation before entering the liver and the concentration exiting the liver: ($C_{in} - C_{out}$)

The extraction ratio, E, is the fraction of drug that is metabolized – extracted – by the liver over a single pass through the liver at steady state:

$$E = \frac{C_{in} - C_{out}}{C_{in}}$$

Clearance depends on the blood flow to the organ:

$$Q_H: CL = \frac{Q(C_{in} - C_{out})}{C_{in}} \rightarrow Q \times E$$

rearranging, $E = \frac{CL}{Q}$

Clearance will be proportional to the **unbound fraction** of drug and the drug's **intrinsic clearance** for that drug. Blood flow does not limit elimination.

high E_H ↓ large first-pass effect ↓ low bioavailability	low E_H ↓ small first-pass effect ↓ high bioavailability
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Most drugs fall between the two extremes.

Drugs that have a **high E_H** have a **large first-pass effect** and **low bioavailability** after oral administration. Hepatic clearance will approximate blood flow.

Drugs that have a **low E_H** have a **small first-pass effect** and **high bioavailability** after oral administration.

Incorrect options:

- $CL = (0.693 \times V_d)/t_{1/2}$: This equation models the relationship between clearance, volume of distribution and half-life: $\rightarrow$ rearrange to $t_{1/2} = (0.693 \times V_d)/CL$:
Half-life is a function of both clearance and volume of distribution and reflects the exponential decline of systemic drug concentration during a dosing interval at steady state.
- $F = AUC \text{ (oral)}/AUC \text{ (I.V.)}$: The area under the concentration vs. time curve is proportional to the dose and the extent of bioavailability for drugs with first-order elimination. It represents the total drug exposure across time (from zero to infinity).
AUC reflects the absolute bioavailability (IV, $F=1$) and relative bioavailability of different formulations of one drug, eg, oral, SL, transdermal, IM or subQ, etc..
- $K_a = 1/K_d$: K_d is the equilibrium dissociation constant, which describes drug-receptor interactions at equilibrium. K_d is the concentration at which 50% of receptors are occupied. K_a is the association constant. It is the inverse of K_d .
If K_d is high, meaning a large number of molecules are required to occupy 50% of the receptors, K_a is low, indicating that the drug has a low affinity for the receptors. Conversely, if K_d is low, a relatively smaller number of molecules is required to occupy 50% of the receptors, and K_a is high indicating relatively higher affinity for the receptors.

13. A frail 87-year-old woman with a long history of hypertension and atrial fibrillation develops dyspnea and fatigue of 1-week duration. The patient lives at home alone. A home health aid comes in four days a week for four hours a day. Her son visits every day and her daughter visits once a week.

VS: Pulse 112 bpm, BP 188/72 mmHg, RR 23 breaths per minute, BMI 21 kg/mm².

PE: Jugular venous distention, hepatomegaly, pulmonary crackles, S1, S2, S3 heart sounds, and pedal edema are noted

Labs: Samples for complete blood count, electrolytes, cardiac troponin, BUN, creatinine, and liver function tests are obtained.

Imaging: Chest radiograph shows cardiomegaly and evidence of pulmonary edema. EKG and Doppler echocardiogram are consistent with congestive heart failure with reduced ejection fraction.

The patient is admitted for management of acute decompensated heart failure.

In view of the patient's advanced age, frailty, and clinical condition, what are the potential alterations in the pharmacokinetic properties?

What are the clinical implications for pharmacotherapy in this patient?

Hint: Notes p23-24 (Please try to avoid the temptation to copy and paste the answers.)

Pharmacokinetics changes in the older patient with heart failure	
oral absorption	
volume of distribution	
plasma protein binding	
metabolism (hepatic clearance)	
bioavailability	
renal excretion	
half-life	

Rationale: Alterations in pharmacokinetics should be anticipated.

This patient is elderly and frail and presenting with acute decompensated heart failure. Significant alterations in pharmacokinetics should be anticipated so the patient's drug therapy and monitoring for therapeutic and toxic effects can be adjusted accordingly. Additionally, the patient's nutritional status is unknown at this point.

In addition to the normal changes that occur with aging, heart failure reduces blood flow to the organs, including the liver and kidneys, which reduces metabolism and renal clearance, and mesentery, which may reduce the rate and extent of drug absorption.

Pharmacokinetics changes in the older patient with heart failure	
oral absorption	Heart failure reduces mesenteric blood flow, which slows removal of drugs from the intestinal tract to the systemic circulation. Reduced blood flow can lead to intestinal wall edema, which reduces the extent of absorption
	Other conditions that may alter drug absorption include poor nutrition, slowing of gastric emptying especially diabetics, consuming antacids and/or laxatives. There is little evidence of major alteration in absorption with age.

volume of distribution	Older adults have reduced lean body mass, reduced body water, and increased fat as a percentage of body mass compared to young adults.
	Edema / Ascites / Pleural effusion: the V_d of hydrophilic drugs may be significantly increased.
	Obesity: The V_d of lipophilic drugs may be significantly increased and the V_d of hydrophilic drugs may also be increased because adipose tissue has almost as much water as other tissues.
plasma protein binding	Older patients may have decreased serum albumin which binds many acidic and neutral substances and drugs. The bound to free drug ratio may be significantly altered. Older adults may digest and absorb protein less efficiently and may consume less protein due to reduced appetite, difficulty chewing, or dietary restrictions. Physiologically, older adults may have reduced stomach acid necessary for breaking down protein and slower gastric emptying.
metabolism (hepatic clearance) VERY IMPORTANT	The rate of metabolism of some, but not all, drugs may be slower. The greatest changes occur with reactions by microsome P450 enzymes (CYPs). - Reduced hepatic blood flow may reduce clearance of drugs with high hepatic extraction ratio. <i>Heart failure dramatically reduces liver blood flow.</i> - Patients with a history of liver disease (alcohol, viral hepatitis, metabolic dysfunction-associated steatohepatitis, MASH, formerly named nonalcoholic fatty liver disease) may reduce hepatic metabolism and transport of drugs. - Malnutrition
bioavailability	Reduced hepatic metabolism may increase bioavailability of certain drugs, necessitating dose adjustment or possibly avoidance.
renal excretion (renal clearance) VERY IMPORTANT	the kidney is the major organ of excretion of most drugs. A decline in the estimated glomerular filtration rate (eGFR) occurs with aging in about two-thirds of the population – but is NOT reflected in the serum creatinine value because, as muscle mass declines with age, production of creatinine also declines. Thus, renal clearance declines with age, increasing the half-life and potential for drug accumulation and toxicity.
half-life	Half-life and clearance are inversely proportional – as clearance decreases, half-life increases, which may, again, necessitate dose adjustment.

14. JP is a 7-year-old male diagnosed with focal seizures with secondary generalization. He is initiated on a standard starting dose of phenytoin. His mother is directed to have her son thoroughly chew 1 tablet 3 times daily (Sig: Chew 1 tab TID). He can take it with food if it bothers his stomach. The doctor also talks to JP to make sure he understands why he is getting the medicine and how to take it and that the tablets taste minty.

JP returns to the neurologist in 7 days for evaluation. His phenytoin trough plasma concentration is 6 mcg/mL (therapeutic range 10-20 mcg/mL). He has not had a seizure event during this time and has been taking the medication exactly as directed. The physician wants to double the serum concentration.

If the dose of phenytoin is doubled, what serum concentration could be anticipated?

- A. The C_p will double because of linear elimination kinetics.
- B. The C_p may significantly increase because of nonlinear elimination kinetics.*

Rationale: B. nonlinear elimination

Phenytoin exhibits zero-order and nonlinear elimination at therapeutic concentrations.

➤ **Practice point:** Therapeutic drug monitoring can guide dosing as the phenytoin dose is gradually increased.

Differentiating the Terms:

First-order and zero-order elimination refer to the *rate of elimination*.

Linear and nonlinear elimination refer to the *plasma concentration v. time relationship*.

Feature	First-order elimination Linear elimination	Zero-order elimination Nonlinear elimination
Elimination rate	Proportional to drug concentration Linearity: Doubling the dose will double the plasma concentration	Constant, independent of concentration Nonlinearity: Small dose increases can lead to large changes in plasma concentrations
Saturation of elimination systems	Elimination systems are not saturated	Elimination systems are saturated (at concentrations $>K_m$)
Half-life	Constant and predictable	Variable and unreliable
Toxicity risk	Dose increase → proportional increase in elimination → elimination processes are efficient and adaptable → lower potential for toxicity	Constant elimination rate independent of drug concentration → potential for drug accumulation over time → higher potential for toxicity
Predictability	High predictability Easier to predict clearance and adjust dosage	Low predictability Requires careful monitoring and individualization of dosing

Details on zero-order and nonlinear elimination:

Zero-order elimination occurs when the rate of reaction is *constant and independent* of the concentrations of the reactants:

- Drug is eliminated at a constant rate regardless of the concentration of drug in the body.
- The same amount of drug (eg, mg/L) will be eliminated per unit of time (eg, per hour)

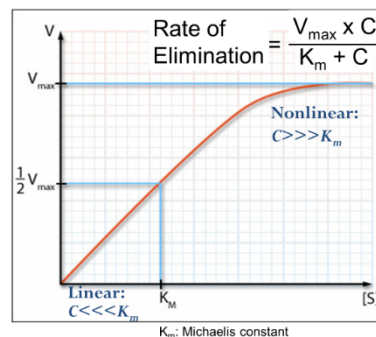
Nonlinear elimination: If the dosing rate exceeds the drug-metabolizing capacity due to saturation of the enzymes, the drug can accumulate.

That is, small dose increases can result in large increases in serum concentration. Thus, *it is difficult to anticipate the plasma concentration when the dose of phenytoin is increased.*

Why half-life does not apply in zero-order elimination:

Since a constant amount of drug is removed per unit of time, the concept of having a fixed time interval (half-life) does not apply. The time to decrease the plasma concentration by half changes as drug concentration increases or decreases.

NOTE: The graph shows the Michaelis-Menten saturation curve showing the relation between the substrate concentration (x axis) and the reaction rate velocity (V) (y axis).



Zero-order kinetics

$$C = \frac{\text{Dosing rate} \times K_m}{V_{max} - \text{Dosing rate}}$$

As dosing rate approaches V_{max} , ($V_{max} - \text{dosing rate}$) approaches zero → disproportionate ↑ C
At $C \gg K_m$ elimination rate is almost independent of concentration

Goodman & Gilman's The Pharmacological Basis of Therapeutics, 12e, 2011; Chapter 2: Pharmacokinetics > Clinical Pharmacokinetics

STUDENTS: EXPLANATION OF THE ScholarRx chart

In past years, students have asked:

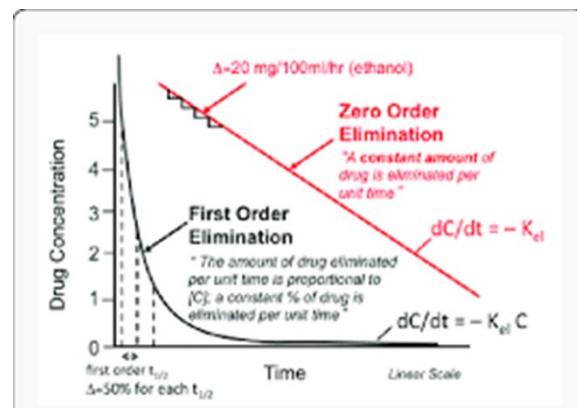
Why is zero-order elimination nonlinear when the concentration-time curve is a straight line?

And

Why is first-order elimination linear when the concentration-time curve is a curved line?

Answer:

- ❖ Zero-order kinetics is a straight line on the concentration-time curve because a constant amount of drug is eliminated.
 - Metabolic enzymes / renal transporters are saturated.
 - The chart is showing that, for every 100 mL of blood, ethanol is eliminated at a constant rate of 20 mg per hour.



Recap:

The rate of drug elimination is fixed → straight line

Nonlinear elimination because small increases in dose can cause large changes in plasma concentration

- ❖ First-order kinetics is a curved line because the elimination rate is directly proportional to the drug concentration
 - Elimination routes (liver / kidneys) are not saturated.
 - The change in plasma concentration is 50% for every half-life
 - Example: 50% of the plasma concentration is eliminated per unit of time
 - If the initial plasma concentration is 100 mg/L,
 - one $t_{1/2} \rightarrow 50$ mg/mL
 - two $t_{1/2} \rightarrow 25$ mg/mL
 - three $t_{1/2} \rightarrow 12.5$ mg/mL
 - and so on until drug is eliminated from the body

Recap:

The elimination rate is directly proportional to plasma concentration $\rightarrow$ curved line

Linear elimination $\rightarrow$ doubling the dose will double the plasma concentration of the drug because elimination mechanisms are not saturated and can increase in activity to clear the drug at a faster rate.

Directly proportional means:

- As the plasma concentration decreases, the rate of elimination decreases – a smaller amount of drug is eliminated over time
- Conversely, as the plasma concentration increases, the rate of elimination increases until the amount of drug administered = the amount of drug eliminated (steady state)

15. A hospitalized patient is treated for supraventricular arrhythmias with verapamil continuous I.V. infusion at 10 mg/h. When stable, she is switched to an oral extended-release verapamil capsule at 1 cap every 24 hours. The patient's weight is 70 kg.

Which one of the following calculations would be useful in predicting the dose of a once daily extended-release capsule to maintain the drug concentration within the therapeutic window?

A. $\frac{(CL \times C_p) \times 24 \text{ h}^*}{F}$	B. $\frac{(V_d \times C_p) \times 24 \text{ h}}{F}$	C. $\frac{(CL \times AUC) \times 24 \text{ h}}{F}$	D. $\frac{24 \text{ h AUC-IV}}{24 \text{ h AUC-oral}}$
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Rationale: A. $\frac{CL \times C_p}{F} \times 24 \text{ h}$ maintenance dose calculation

Bioavailability, F , is factored in to account for first-pass metabolism with oral dosing.

The principle of maintenance dosing is to maintain the drug within the therapeutic window by giving just enough drug in each dosing interval to replace the drug eliminated since the preceding dose: the rate of drug in = the rate of drug out.

Recall: Steady state is achieved when the rate of elimination equals the rate of drug administration.

Incorrect options:

- B. $(V_d \times C_p) / F \times 24 \text{ h}$ is the formula for calculating loading dose.
- C. $(CL \times AUC) / F \times 24 \text{ h}$ is a rearrangement of a clearance calculation: $CL = \text{dose}/AUC$. Bioavailability determines the AUC – the total body exposure to the drug over a given time period. F is not included in the clearance calculation using this method. F is in the denominator of this option is to keep the options parallel to each other.
- D. $AUC\text{-IV} / AUC\text{-oral}$ is the formula for determining the relative bioavailability of an oral dosage form of a given drug to the intravenous form of the same drug. 24 h AUC doesn't necessarily apply in this context. AUC – total drug exposure over time – is expressed as the integral from 0 to ∞ . The frequency of blood collection when comparing the AUC of the IV form to another form of the same drug, or different manufacturers of the same drug and formulation, aims to capture the absorption phase (where applicable), distribution phase, and elimination phases. Duration continues until adequate data is captured.

16. A 44-year-old male presents to the ED with a blood pressure of 185/112 mmHg. He is treated with an I.V. antihypertensive agent.

Which of the following is the preferred method of monitoring the effect of this drug?

- A. Blood pressure measurement*
- B. Cardiac telemetry
- C. EKG
- D. Therapeutic blood levels

Rationale: The therapeutic goal of antihypertensive drugs is lowering blood pressure. Monitoring effect is easily done by taking blood pressures.

Incorrection options:

Cardiac telemetry and EKG are useful for monitoring electrical signals of the heart.

Measuring drug plasma levels (therapeutic drug monitoring) is useful for optimizing therapy of drugs with:

- narrow therapeutic windows and potential for toxicity,
- wide inter- and intra-patient pharmacokinetic variability,
- nonlinear kinetics, pregnancy,

and in for patients with disease states or very young and very old ages that affect the pharmacokinetics of the drugs.

- **Practice point:** Therapeutic drug monitoring can only be done for drugs having *therapeutic drug assays* and when there are *clear correlations* between serum concentrations and efficacy or tolerability.

17. A generic drug manufacturer is developing a version of a popular brand-name antihypertensive tablet, whose patent has expired. The company submits to the FDA data demonstrating that:

- the drug contains the identical active ingredient in identical amounts as the reference drug,
- the rate and extent of absorption do not show a significant difference from the rate and extent of absorption of the reference drug, and
- the generic version can be expected to have the same clinical effect and safety profile as the reference drug when administered to patients under conditions specified in the labeling.

Which one of the following is the generic drug maker demonstrating to the FDA?

- A. The generic version is a bioequivalent.
- B. The generic version is a pharmaceutical equivalent.
- C. The generic version is a therapeutic equivalent.*
- D. The generic version is a biosimilar.
- E. The generic version is an interchangeable biological.

Rationale: C. therapeutic equivalence – the one best answer

According to FDA terminology, therapeutic equivalence must be pharmaceutically equivalent, bioequivalent, and have the same clinical effect and safety profile as the comparator drug.

The generic product defined in the vignette shows bioequivalence and it can be expected to have the same clinical effect and safety profile as the reference drug.

Incorrect options:

- Bioequivalence: Two pharmaceutically equivalent drug products are considered to be *bioequivalent* when the rates and extents of bioavailability of the active ingredient in the two products are not significantly different under suitable test conditions.
- Pharmaceutical equivalence: Drug products that contain the same active ingredients and are identical in strength or concentration, dosage form, and route of administration
- Biosimilar: Biological product highly similar to an FDA-approved biological product and having no clinically meaningful differences in safety and effectiveness from the reference product
- Interchangeable biological: A biosimilar product that meets additional standards for interchangeability, and which may be substituted for the reference product by a pharmacist without the intervention of the health care provider who prescribed the reference product

FDA Orange Book Preface, (3/27/2025) <https://www.fda.gov/drugs/development-approval-process-drugs/orange-book-preface>

18. A pharmaceutical company is developing a mint-flavored orally disintegrating tablet of their popular antihistamine oral tablet. It rapidly dissolves when placed on the tongue and can be taken without water.

Which of the following parameters is most useful in comparing both the rates and extents of bioavailability of the company's oral tablet and the orally disintegrating tablet of their drug?

- A. AUC*
- B. Bioavailability
- C. CL
- D. $t_{1/2}$
- E. V_d

Rationale:

AUC area under the concentration-time curve is used to compare relative bioavailabilities of the two formulations:

$$F = \frac{AUC_{\text{formulation-A}}}{AUC_{\text{formulation-B}}}$$

AUC provides a measure of the overall amount of drug that has reached the systemic circulation and been present in the body over time – ***the total body exposure to the drug.***

- The AUC is directly proportional to the dose and the extent of bioavailability for drugs with first-order/linear elimination.
- AUC is inversely proportional to the clearance: $\frac{\text{dose in mg} \times F}{\text{CL in L/h}}$ (oral drugs)

The higher the clearance, the lower the AUC.

The lower the clearance, the higher the AUC.

- AUC units: mass * time / volume → mg*h/L
For example: AUC = 300 mg*h/L

Bioavailability
+
Time
↓
AUC

Clinical relevance:

Single-dose AUC: Early drug development for determining initial dosing recommendations

Steady-state AUC: Long-term therapy for chronic disease to ensure therapeutic levels are maintained without accumulation or toxicity, especially for drugs with narrow therapeutic window.

Katzung's Figure 3-4

Blood concentration-time curves illustrating how changes in the rate of absorption and extent of bioavailability can influence both the duration of action and the effectiveness of the same total dose of a drug administered in three different formulations.

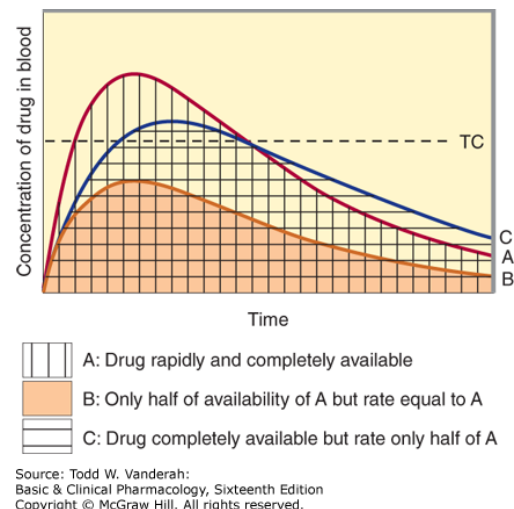
The dashed line indicates the target concentration (TC) of drug in blood.

Intravenous dose: bioavailability = 100%

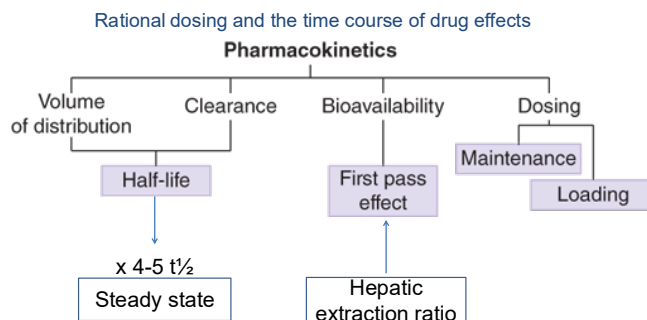
Oral dose: bioavailability may be <100% due to:

- 1) Incomplete absorption from the GI tract, across the gut wall, into the portal circulation, and
- 2) First-pass elimination

Source: Access Medicine Katzung Basic & Clinical Pharmacology, 14e, 2018; Chapter 3: Pharmacokinetics & Pharmacodynamics: Rational Drug Dosing & the Time Course of Drug Action; Figure 3-4



SUMMARY



- The plasma concentration is a function of the rate of input of the drug (by absorption) into the plasma, the rate of distribution into other tissues, and the rate of elimination.
- If the rate of input is known, the remaining processes are well described by two primary parameters: **apparent volume of distribution (V_a)** and **clearance (CL)**. These parameters are unique for a particular drug and a particular patient but have average values in large populations that can be used to predict drug concentrations.
- Clearance of a drug is the factor that predicts the rate of elimination in relation to the drug concentration.
- Volume of distribution is the proportionality constant that relates the total amount of drug in the body to the plasma concentration at a given time.
- Change in blood flow to various tissues change in rates of drug distribution to tissues. Thus, the effect of a drug at various sites of action can vary depending on perfusion of these sites.
- Removal of a drug by an organ can be specified as the extraction ratio – the fraction or percentage of the drug removed from the perfusing blood during one passage through the organ.
- The bioavailability of a drug is the fraction (F) of the administered dose that reaches the systemic circulation.

- The area under the concentration vs. time curve (AUC) describes the measured concentration of drug in the systemic circulation as a function of time (from zero to infinity) – the concentration of drug appearing in plasma occurs over time. It can be used to calculate bioavailability.
- The elimination half-life is a measure of the time it takes for the plasma concentration or the amount of drug in the body to be reduced by 50%. Half-life is a function of volume of distribution and clearance. Half-life elimination curves track the amount of drug in the body over time.
- In first-order elimination kinetics, the drug elimination rate is directly proportional to plasma free drug concentration. Most clinically relevant drugs follow first-order kinetics.
- In zero-order elimination kinetics, the metabolic enzymes eventually will become saturated as the concentration of substrate increases. The rate of drug elimination is constant (a constant amount) and independent of plasma drug concentration. A few drugs follow zero-order kinetics at therapeutic concentrations.
- When a drug that exhibits first-order pharmacokinetics is administered to a patient continuously or intermittently, the drug will accumulate until it reaches a plateau – a steady-state – plasma drug concentration.
- The time to reach the steady state is a function of the elimination half-life of the drug. It takes 4 to 5 half-lives to approach steady state. If the half-life or the dosing rate changes, it will again take 4-5 half-lives to reach the new steady state plasma concentration.
- The therapeutic goal of the target concentration strategy is to maintain steady state drug levels within the therapeutic range to provide therapeutic efficacy and minimum toxicity. This approach links pharmacokinetics and pharmacodynamics.
- A loading dose rapidly raises the plasma drug concentration to the target concentration. The loading dose is the total amount of drug in the body immediately following the loading dose. Volume of distribution relates the total amount of drug in the body to plasma drug concentration:
- $LD = (V_d \times C_{desired}) / F$
- The maintenance dose is administered after the loading dose to maintain desired steady state concentration. That is, the amount given replaces the amount that is eliminated since to preceding dose – rate in = rate out: $MD = (CL \times C_{desired}) / F$.
- Disease states may modify pharmacokinetic parameters and pharmacodynamic actions and response to drug therapy.
- Optimization is based on in-depth understanding of factors that determine an individual's response to drug treatment.
- *Fundamental principles should guide prescribing of drugs.*

Resources:

Access Medicine Goodman & Gilman's The Pharmaceutical Basis of Therapeutics, 14e, 2023;
Chapter 2: Pharmacokinetics: The Dynamics of Drug Absorption, Distribution, Metabolism, and Elimination

Access Medicine Katzung's Basic & Clinical Pharmacology, 16e, 2024; Chapter 3:
Pharmacokinetics & Pharmacodynamics: Rational Drug Dosing & the Time Course of Drug
Action; Chapter 60: Special Aspects of Geriatric Pharmacology
Lexidrugs monographs
LWW Health Library Medical Education: Lippincott Illustrated Reviews: Pharmacology; 7e,
2017; Chapter 1: Pharmacokinetics